

Technical Assignment: Build an Agentic RAG Application

Objective

Build an Agentic RAG application that enables users to upload documents, interact through a chat interface, and receive accurate answers with citations including page numbers from the original sources.

- Chat Assistant - Upload documents	<i>Answer</i> _____ _____ _____ _____ Pages: [1, 2, 3] Documents: - filename.pdf - filename2.doc Input: Ask question Send
---	--

Requirements

Core Features

- **Chat Interface:**
 - Implement an **interactive interface**.
 - The interface should be **simple and functional** (design is not a priority).
 - **Question Answering:**
 - Users upload documents and ask questions.
 - The system should:
 - Provide answers based on uploaded documents.
 - Include file name and page number as citation.
 - If no answer exists, respond clearly with “No answer found” (or similar).
 - Avoid AI hallucinations.
 - **Core Components:**
 - Use **LlamaIndex** framework for retrieval and indexing (mandatory).
 - Use any **Vector Database** for storing embeddings, You may choose any DB you prefer (e.g., Neo4j, Weaviate, etc).
 - **Optional Components:**
 - **LlamaParse** for document parsing and extraction.
 - Additional tools, frameworks, or APIs that enhance performance or user experience.
 - **Agentic Behaviour** (Optional):
 - Support **multi-step reasoning** for complex queries.
 - **Conversation Memory:**
 - Support memory so the system can understand **follow-up questions** in context.
-

Technical Specifications

- **Language:** Python
 - **Main Tools:** LlamaIndex + (You can use additional libraries or tools if they improve functionality).
 - **Containerization:** Docker
 - Deploying the app on a **free cloud platform** (like Streamlit Cloud, Render, or similar) if possible (read deliverables part).
-

Deliverables

- **GitHub Repository** (or compressed folder) including:
 - Source code.
 - **Dockerfile** and configurations for deployment.
 - **README** with:
 - Setup and run instructions.
 - Architecture overview.
 - Environment variables and configuration details.
 - **Optional:** If deployment is not possible, a short screen recording (max 5 minutes) showing the app in action.
 - Include the total duration (how many hours/days you spent on the task).
-

Test Data

We will provide:

- A set of **PDF documents**
- A **list of questions with expected answers** for evaluation.

Additional Notes on Evaluation

- A list of **sample questions with expected answers** will be provided for testing.
- These questions and answers serve as **guidelines**, not strict benchmarks.
 - Your answers should be **correct and sourced from the documents**.
 - They **do not need to match the expected answer word-for-word**, but the **accuracy and precision** of the response matter.
 - Avoid fabricating or guessing. If no relevant information exists, return **“No answer found”**.